Manish Kumar

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Professional Summary

Software Engineer specializing in AI/LLM applications, agentic systems, real-time voice pipelines, and enterprise automation. Experienced in building scalable, production-ready systems using Python, FastAPI, React Native, Gemini, Groq, Deepgram, WebSockets, and Docker with a focus on sub-second latency, self-healing architectures, and reliable deployment.

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++, SQL, HTML, CSS
AI & Retrieval: Multi-agent systems, retrieval-augmented generation (RAG), ChromaDB, Azure OpenAI embeddings, Gemini, Groq, tool calling
Speech & Audio: Deepgram Nova-3 (speech recognition), Aura (speech synthesis), OpenSpline avatar animation, Web Audio API, AudioWorklet
Backend & Data: FastAPI, WebSockets, asyncio, Firebase Cloud Firestore, Vercel Serverless Functions
Web & Mobile: React, React Native, Expo, Tailwind CSS, Streamlit, UPI deep links
Deployment & Tools: Docker, Render, Vercel, Expo EAS Build, Git, GitHub, Linux

Experience

Associate Software Engineer, Accenture Aug 2026 – Present

- Developing AI-driven automation for enterprise IT operations, streamlining ServiceNow workflows and reducing repetitive manual ticket triage.
- Integrating AI/ML capabilities with enterprise IT service-management systems to improve ticket routing, categorization, and operational productivity.
- Building scalable and maintainable backend software components for enterprise automation, emphasizing production reliability and performance.

Software Developer Intern, Accenture Dec 2025 – May 2026

- Developed intelligent automation components for enterprise IT operations, optimizing ticket resolution workflows and reducing manual intervention.
- Integrated AI/ML logic with IT service-management systems to accelerate triage and route incident escalations with high accuracy.

Projects

Speak with Sway – Live AI English Tutor [Live] [GitHub]

- Built a spoken English tutor that gives phrasing feedback and asks follow-up questions using Deepgram speech recognition and synthesis, Groq, and FastAPI.
- Integrated OpenSpline to synchronize the tutor's facial movements with its voice. Learners can interrupt a reply and continue the conversation.
- Streamed audio over WebSockets with a bounded buffer to keep playback smooth and preserve packet order.

Autonomous IT Operations Agent – RAG & Self-Healing Infrastructure [Live] [GitHub]

- Built a multi-agent RAG system using ChromaDB and Azure OpenAI embeddings to diagnose live server anomalies using historical IT runbooks.
- Developed Diagnoser and Planner agents with tool calling and a Human-in-the-Loop (HITL) authorization gate for controlled remediation.
- Built a Streamlit Mission Control dashboard with incident simulation, knowledge-base exploration, and audit logging for autonomous self-healing workflows.

Apna Kirana – Hyperlocal Commerce & Store Management Engine [Live Demo] [GitHub]

- Built a commission-free hyperlocal ordering platform with a React Native (Expo) customer app and a real-time web POS for store counters.
- Engineered a Hindi Voice Soundbox announcing orders aloud in real time, with instant UPI deep-link checkout and sub-second status sync.

Education

Rajiv Gandhi Institute of Petroleum Technology 2022 – 2026
B.Tech. in Computer Science and Engineering CPI: 7.69
Dr. I.S.N.R.K.P.L.D. Kisan College, Sitamarhi – Intermediate: 80.8% 2019 – 2021
Simultala Awasiya Vidyalaya – Matriculation: 94.2%, Bihar Board State Rank 13 2018 – 2019

Achievements

- **Winner**, IIT Mandi Hackathon | **Grand Finalist**, IIT Bombay Hackathon
- **JEE Advanced Top 2%** | **Bihar Board State Rank 13** in Matriculation | **NTSE Stage 1 Qualified**